

ACDB v.1 codebook*

Table 1. Definitions for variables at the species level

species	
species_id [primary key]	Scientific name of the organism(s) in the format Genus_species. If the population/behavior was attributed to a subspecies, formatted as Genus_species_epithet. If the source(s) describing the group or behavior used an older version of the species name, the accepted form of the species name in GBIF is used
common_name	Common name of the species in English
GBIF	GBIF usageKey for the species following gbif.org/species/
canonicalName	Canonical name from GBIF API
Genus ...Phylum	Taxonomic names from GBIF backbone
primary_social_unit unit_evidence unit_source	The largest social unit for this species in which individuals maintain stable membership, adapted from Prox & Farine (2020) and Groot et al. (2023). Individual (no stable association with conspecifics); family (unit is composed of mating pair/related individuals); group (unit includes unrelated individuals, may display fission-fusion behavior)
IUCN	Current IUCN Red List status

Table 2. Definitions for variables on social groups within species to which cultural behaviors pertain

groups	
group_id [primary key]	Unique ID for each group. Integer starting with 700000

species_id	Foreign key linking to species table
group_name	Name of the group or population, e.g. Northern Resident killer whales or Gombe chimpanzees. Assigned based on location if none appears to be commonly used
group_level	The group level, relative to the species primary unit, to which cultural behaviors are attributed. 0+ with primary unit as 0; e.g. for killer whales 0 = matriline, 1 = pod, 2 = clan, 3 = community/population, 4 = ecotype, 5 = species
group_above	In the case of species with nested social structure, name of group directly above or encompassing the coded group; e.g. if focal group is Southern Resident killer whales, the group above this is the resident killer whale ecotype
size size_evidence - size_date - size_source -	Approximate group size in number of individuals
location_evidence	As specific information possible on location, including research site, city, country, etc.
lat	Approximate latitude for location of population
long	Approximate longitude for location of population

Table 3. Definitions of variables describing behaviors for groups.

behaviors	
group_id	Foreign key linking to groups table
behavior_id [primary key]	Unique ID for each behavior. Integer starting

	with 2
behavior	Short name describing behavior, e.g. vocal dialect, termite fishing, lobtail feeding
behavior_description	Description of the behavior, placed in context of the species and the focal population(s)
behavior_source	Primary source used to describe the behavior; links to sources table
start_date	First observed date of behavior in the literature
end_date	Most recent observed date of behavior in the literature
vertical vertical_evidence - vertical_source -	Behavior is transmitted from parents to offspring. yes, inferred yes, unclear (the primary researchers presumably know but don't mention it), maybe (the primary researchers speculate about it), inferred no, no (evidence against), suspected unknown (no research on transmission exists)
horizontal horizontal_evidence - horizontal_source -	Behavior is transmitted among peers in the same generation
oblique oblique_evidence - oblique_source	Behavior is transmitted from non-parent adults to juveniles
domains domains_evidence - domains_source	One of more of migration (knowledge of migration routes or sites); foraging (methods or tools of hunting or obtaining food, knowledge of food or resource patches, etc.); vocal communication (vocal transmission of information to others, e.g. songs and vocal dialects); antipredation (predator knowledge and avoidance); play (nonfunctional behavior that appears to be 'for fun'); mating (mating displays, breeding sites, mate selection); architecture (modifications to the physical environment- nest building, burrows, etc.);

	social (non-mating social behaviors, such as those that reinforce group identity)
anth_effects anth_source -	Any effects of human activity on the behavior

Table 4. Metadata for variable sources and references.

sources	
source_id [primary key]	Unique ID for each source in format firstauthoryear, e.g. baker2004
title - year - authors - doi -	Title of article or book chapter, year of publication, all authors listed as text, DOI
methods	Description of the methods used to establish the presence of social learning and the existence of cultural behavior in the study population(s), following as closely as possible Whiten & Rutz (2025)

*Planned additions:

- Species table: Dispersal/migration
- Behaviors: IUCN threats- Anthropogenic threats to the behavior coded according to the IUCN–CMP Unified Threats Classification Scheme (<https://www.iucnredlist.org/resources/threat-classification-scheme>); Sex-biased transmission; Observation time